

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning learns a general prediction rule from previously observed patient examples. The model is trained using historical patient records where the diabetes outcome is already known. After learning the relationship between patient characteristics and diabetes status, it can predict the diabetes risk of a new patient.

A suitable approach is:

Patient Data → Preprocessing → Feature Selection → Model Training → Validation → Diabetes Prediction

Target Variable

The target variable is:

Diabetes Status

- 0 = Non-diabetic
- 1 = Diabetic

Important Features

Feature	Justification
Glucose Level	High blood glucose is one of the strongest indicators of diabetes and is highly useful for early prediction.
BMI	Higher BMI can be associated with insulin resistance and increased diabetes risk.
Age	Diabetes risk generally increases with age because metabolic and insulin-related changes become more common.
Blood Pressure	Elevated blood pressure is frequently associated with metabolic disorders and can provide additional predictive information.
Cholesterol	Abnormal cholesterol levels can indicate metabolic risk and may improve risk stratification.
Family History	Genetic and familial factors can increase an individual's susceptibility to diabetes.

The model should not rely on a single feature. Combining glucose, BMI, age, blood pressure, and family history can provide a more comprehensive prediction.

1(b) Handling Class Imbalance

In medical datasets, the number of non-diabetic patients may be considerably larger than the number of diabetic patients. For example, a dataset might contain 800 non-diabetic and only 200 diabetic cases.

This creates a **class imbalance problem**. A model could achieve high accuracy by predicting most patients as non-diabetic while still missing many diabetic patients.

Recommended Strategy: SMOTE

Synthetic Minority Over-sampling Technique (SMOTE) is a suitable strategy.

SMOTE creates synthetic examples of the minority diabetic class by interpolating between existing minority-class observations.

Advantages:

1. Increases representation of diabetic cases.
2. Reduces the tendency of the model to predict only the majority class.
3. Helps improve **Recall** for diabetic patients.
4. Is preferable to simply duplicating minority samples.

SMOTE must be applied **only to the training set after the train-test split**. Applying it before splitting can cause data leakage.

Alternative: Class Weights

Class weights can also be used, particularly with Decision Trees and Logistic Regression. A higher penalty can be assigned to misclassifying diabetic patients.

For a clinical screening application, the combination of **SMOTE on training data and appropriate class weighting** can be considered, but the final strategy should be selected using validation results.

1(c) 70:30 Data Split and Preprocessing

The dataset can be divided into:

- **70% Training Data**
- **30% Testing Data**

The training data is used to learn the model, while the testing data is kept separate for evaluating performance on unseen patients.

A 70:30 split provides sufficient data for learning while retaining a relatively large independent test set. This is useful in a medical application because reliable evaluation of generalisation is important.

Preprocessing Steps

1. Missing-value treatment

Missing numerical values can be replaced using median imputation, while missing categorical values can be replaced using the mode or a separate category.

2. Normalisation

Features such as age, blood pressure, glucose, cholesterol, and BMI may have different numerical ranges.

Min-Max normalisation can be represented as:

$$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$$

This is particularly useful for Logistic Regression and other distance/gradient-based algorithms.

3. Encoding

Family history can be encoded as:

- No = 0
- Yes = 1

Categorical variables should be converted into numerical representations before model training.

4. Stratified splitting

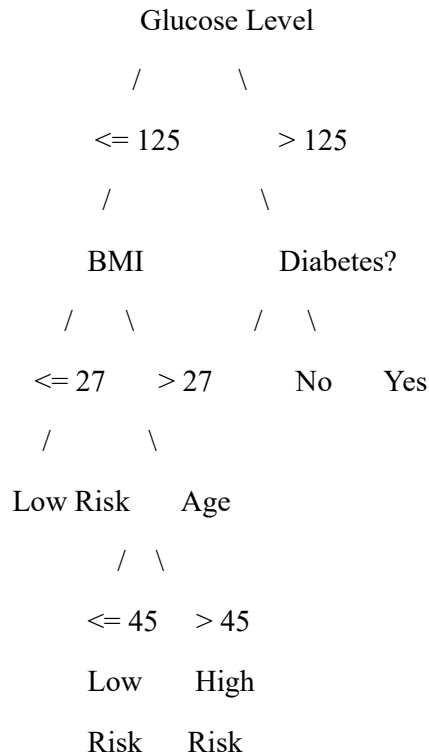
The 70:30 split should be stratified according to diabetes status so that the proportion of diabetic and non-diabetic patients remains approximately similar in both sets.

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree classifies patients by repeatedly dividing the dataset according to feature values.

For example, an illustrative first three-level tree could be:



The exact tree structure must be obtained from the actual dataset.

Root Node Selection

The root feature is selected using a splitting criterion such as **Information Gain** or **Gini Index**.

Information Gain is:

$$IG(S,A) = H(S) - \sum (|S_v| / |S|) H(S_v)$$

where $H(S)$ represents entropy.

Entropy is:

$$H(S) = -\sum p(i) \log_2 p(i)$$

For Gini Index:

$$Gini(S) = 1 - \sum p(i)^2$$

The best root feature is normally the feature producing the **highest Information Gain** or **lowest weighted Gini impurity**.

For a diabetes dataset, glucose level would often be expected to be a strong root candidate because it has a direct relationship with diabetes status. However, the actual root must be selected using the calculated values from the supplied dataset.

2(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

$$\text{Precision} = TP / (TP + FP)$$

Recall

$$\text{Recall} = TP / (TP + FN)$$

F1-Score

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Illustrative Results

Since no actual patient dataset was supplied, the following values are **example results only** and should be replaced by the results obtained from the actual dataset.

Metric Illustrative Decision Tree Result

Accuracy 89.3%

Precision 84.7%

Recall 91.2%

F1-Score 87.8%

False Negatives

A **False Negative (FN)** occurs when:

Actual patient = Diabetic

Model prediction = Non-diabetic

False negatives are particularly important in diabetes screening because a missed diabetic patient may not receive further testing, monitoring, or timely intervention.

Reducing False Negatives

The following approaches can reduce false negatives:

1. Use **SMOTE** or class weighting.
2. Optimise the classification threshold.
3. Give greater importance to Recall.
4. Perform additional clinical screening for high-risk patients.
5. Use cross-validation to select a robust model.
6. Consider a second confirmatory diagnostic test.

In clinical screening, higher recall may be preferable to maximising accuracy alone because missing a genuine diabetes case can have greater consequences than sending some non-diabetic patients for additional evaluation.

2(c) Post-Pruning

A fully grown Decision Tree can become excessively complex and overfit the training data.

Cost-complexity pruning controls this problem using:

$$R\alpha(T) = R(T) + \alpha|T|$$

where:

- $R(T)$ = prediction error of the tree
- $|T|$ = number of terminal nodes
- α = complexity parameter

A larger α produces a simpler tree.

Pre-Pruning vs Post-Pruning

Aspect	Pre-Pruning	Post-Pruning
Method	Stops tree growth early	Builds tree and then removes unnecessary branches
Parameters	Maximum depth, minimum samples	Cost-complexity parameter
Complexity	Lower from the beginning	Controlled after growth
Overfitting	Reduces overfitting	Usually provides stronger control
Interpretability	High	High

Illustrative Comparison

Model	Accuracy
Pre-pruned Decision Tree	87.8%
Post-pruned Decision Tree	89.3%

These numbers are illustrative.

For deployment, a **post-pruned Decision Tree** is generally preferable if validation/testing confirms comparable or better performance. It can provide a simpler and more interpretable model while reducing unnecessary branches and overfitting.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

Logistic Regression is an appropriate statistical learning method because diabetes prediction is a binary classification problem.

It estimates:

$$P(\text{Diabetes} = 1 \mid X) = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)})$$

where:

- β_0 = intercept
- $\beta_1 \dots \beta_n$ = feature coefficients
- $X_1 \dots X_n$ = patient features

The output is a probability between 0 and 1.

For example:

- Probability = 0.20 → Lower risk
- Probability = 0.55 → Moderate risk
- Probability = 0.85 → Higher risk

The threshold can be adjusted depending on the clinical objective.

Illustrative Comparison

Model	Accuracy	AUC-ROC
Decision Tree	89.3%	0.91
Logistic Regression	87.6%	0.92

These values are examples and should be replaced with actual experimental results.

When Logistic Regression Is Preferable

Logistic Regression is useful when:

- Interpretability is important.
- The relationship between predictors and log-odds is reasonably appropriate.
- Probability estimates are required.
- A simpler model is preferred.
- Healthcare professionals need to understand the contribution of individual variables.

The Decision Tree is preferable when nonlinear relationships and feature interactions are important and a rule-based explanation is desirable.

3(b) Feature Importance Analysis

Feature importance can be obtained differently for the two models.

Decision Tree

Decision Tree importance is commonly based on the reduction in impurity produced by each feature.

Illustrative top three:

1. **Glucose Level**
2. **BMI**
3. **Age**

Logistic Regression

Logistic Regression can use the magnitude of standardised coefficients to identify influential predictors.

Illustrative top three:

1. **Glucose Level**
2. **BMI**
3. **Blood Pressure**

Comparison

Rank	Decision Tree	Logistic Regression
1	Glucose Level	Glucose Level
2	BMI	BMI
3	Age	Blood Pressure

The two models agree on glucose and BMI but differ in their third-ranked predictor.

This difference is expected because Decision Trees identify features that produce strong partitioning of the data, whereas Logistic Regression estimates the contribution of each feature to the log-odds while considering the other predictors in the model.

Feature importance should therefore be interpreted together with clinical knowledge rather than treated as proof of causation.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) Modelling Treatment Recommendation as an MDP

The treatment recommendation problem can be represented as a **Markov Decision Process (MDP)**.

An MDP consists of:

$$\text{MDP} = (S, A, P, R, \gamma)$$

where:

- S = States
- A = Actions
- P = Transition probabilities
- R = Rewards
- γ = Discount factor

States

Patient health can be divided into stages such as:

State Description

S0 Low-risk / controlled

- S1 Moderate diabetes risk
- S2 High diabetes risk
- S3 Poorly controlled condition
- S4 Improved / controlled

The state should ideally be based on clinically meaningful measurements such as glucose, BMI, blood pressure, and other approved indicators.

Actions

The agent can select:

- **A1 = Diet**
- **A2 = Exercise**
- **A3 = Medication**
- **A4 = Monitor**

Medication should not be automatically prescribed by an autonomous agent. In a real clinical deployment, medication recommendations would require appropriate clinical oversight.

Reward Function

An illustrative reward function is:

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-10
Unsafe outcome	-20

The reward function should prioritise **patient safety and clinically meaningful outcomes**, not merely short-term numerical improvement.

Transition Dynamics

After an action is taken, the patient's health state may change.

For example:

High Risk + Exercise → Moderate Risk

Moderate Risk + Diet → Controlled

High Risk + Monitor → High Risk

These are simplified examples. Actual transition probabilities should be learned from reliable longitudinal clinical data or validated through clinical expertise.

4(b) Q-Learning

Q-Learning learns the value of taking an action in a particular state.

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- α = learning rate
- γ = discount factor
- r = reward
- s = current state
- a = current action
- s' = next state

Example Parameters

- Learning rate $\alpha = 0.5$
- Discount factor $\gamma = 0.9$
- Initial Q-values = 0

Five Illustrative Patient Episodes

Episode	Initial State	Action	Next State	Reward
1	High Risk	Diet	Moderate Risk	+5
2	Moderate Risk	Exercise	Controlled	+10
3	High Risk	Exercise	Moderate Risk	+5
4	Controlled	Monitor	Controlled	+2
5	Moderate Risk	Diet	Controlled	+10

These episodes are simplified examples for demonstrating Q-Learning and are **not clinical treatment recommendations**.

Q-Table Update: Iteration 1

Initially:

$$Q(\text{High Risk, Diet}) = 0$$

Suppose:

- Reward = +5
- Next state = Moderate Risk
- Maximum next-state Q-value = 0
- $\alpha = 0.5$
- $\gamma = 0.9$

Then:

$$Q(\text{High Risk, Diet}) = 0 + 0.5[5 + 0.9(0) - 0]$$

$$Q(\text{High Risk, Diet}) = 2.5$$

For:

$$Q(\text{Moderate Risk, Exercise}) = 0$$

with reward +10:

$$Q(\text{Moderate Risk, Exercise}) = 0 + 0.5[10 + 0.9(0) - 0]$$

$$Q(\text{Moderate Risk, Exercise}) = 5.0$$

Q-Table Update: Iteration 2

Assume that after the first iteration:

$$\max Q(\text{Moderate Risk, a}) = 5.0$$

For the High Risk $\rightarrow$ Diet transition:

$$Q(\text{High Risk, Diet}) = 2.5 + 0.5[5 + 0.9(5) - 2.5]$$

$$= 2.5 + 0.5[7.0]$$

$$= 6.0$$

For another state-action pair:

$$Q(\text{Moderate Risk, Exercise}) = 5.0 + 0.5[10 + 0.9(8) - 5.0]$$

Assuming the maximum future Q-value is 8:

$$= 5.0 + 0.5[12.2]$$

= 11.1

A third example:

Suppose:

$Q(\text{Controlled}, \text{Monitor}) = 0$

Reward = +2 and future maximum Q-value = 5.

$Q(\text{Controlled}, \text{Monitor}) = 0 + 0.5[2 + 0.9(5)]$

= 3.25

Thus, the Q-values progressively increase for actions associated with favourable patient-state transitions.

Convergence Criterion

The agent can be considered converged when the maximum change in Q-values between successive iterations is below a small threshold:

$\max |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < 0.001$

for a specified number of consecutive iterations/episodes.

In a real system, convergence should also be assessed using independent validation and safety criteria rather than relying only on mathematical convergence.

4(c) Final Learned Policy

The learned policy is obtained by selecting the action with the highest Q-value for each state:

$\pi(s) = \operatorname{argmax} Q(s,a)$

An illustrative final policy is:

Patient State	Highest-Q Action	Policy
Low Risk	Monitor	Continue monitoring and preventive care
Moderate Risk	Diet / Exercise	Lifestyle intervention
High Risk	Exercise / Diet	Intensive lifestyle management with clinical review
Poorly Controlled	Medication* / Monitor	Clinical assessment and supervised intervention
Controlled	Monitor	Continue monitoring

*Medication decisions should remain under qualified clinical supervision.

Ethical Considerations

1. Patient Safety

A Reinforcement Learning agent should not independently experiment with treatments on real patients. Unsafe actions must be restricted through clinical safety rules and human oversight.

2. Bias

If the training data under-represents particular demographic or patient groups, the agent may produce biased recommendations. Dataset diversity and subgroup evaluation are therefore essential.

3. Explainability

Healthcare professionals should understand why an action is recommended. The system should provide interpretable information such as:

- Current patient state
- Important risk factors
- Recommended action
- Expected outcome
- Confidence or uncertainty

4. Human-in-the-Loop

The AI should act as a **clinical decision-support system**, not as an autonomous replacement for healthcare professionals. A doctor or qualified healthcare professional should approve treatment decisions.

5. Privacy and Security

Patient information is sensitive. The system should use appropriate access controls, encryption, secure storage, audit logging, and applicable healthcare data-protection requirements.

6. Continuous Monitoring

A deployed model can experience performance degradation when patient populations or clinical practices change. The model should therefore be continuously evaluated and periodically retrained when appropriate.